SRI SAIRAM COLLEGE OF ENGINEERING

Sai Leo Nagar, Anekal -562106

A

LAB MANUAL ON

TECHNICAL WRITING USING LATEX

[BCSL456D]

IV SEMESTER

PREPARED

BY

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Dept. of CSE

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

SRI SAIRAM COLLEGE OF ENGINEERING

Sai Leo Nagar, Anekal -562106

2023-2024

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An introduction to LaTeX

What is Latex?

LaTeX is pronounced as '***lah-tech***' or '***lay-tech***.' which is distributed under the LPPL (Latex Project Public License). It is based on TEX, a typesetting language designed for science and math. Both **LATEX** and **TEX** contain a variety of font styles, such as serif, typewriter, and a set of mathematical functions. It is an open-source and powerful document preparation system. It is high-quality typesetting, which can be used for any publishing. It is often used for scientific publications and medium to large technical documents. It is not a word processor or any word document, but it is an efficient way for the publishers and the authors to organize their materials. It gives a unique appearance to the system. It saves the time for designing as well as the efforts used for the documents.

The **Latex system** handles the layout and formatting of the structures using familiar concepts of the section, table, figure, chapter, etc. It follows the philosophy of separate presentation, i.e., separating the writing contents and the visual appearance so that the authors can focus more on the content rather than its appearance. It is named as the standard for the communication and publication of the technical or scientific documents.

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Features of Latex

- Latex is a markup language and free license software. It includes a set of built-in commands.
- It is a mode of mathematical and special symbols.
- LaTeX is generally distributed along with plain **TEX**, i.e., it provides a set of macros for TEX to interpret. The other macros for TEX are Plain TEX, GNU Texinfo, etc.
- LaTeX is not compatible with the GNU (General Public License). It is available on most of the operating systems like UNIX, BSD, Linux, Windows, DOS, etc.

TeX

TeX is a typesetting system or the formatting system written and designed by **Donald Knuth** released in **1978**. It is the most used typographical system. It performs many typesetting tasks, especially Latex. It is popular in mathematics, computer science,

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engineering, physics, statistics, economics, etc. It was developed to produce high-quality output. The books published in TEX are **Addison Wesley, Cambridge University Press, Oxford University Press,** etc. It was primarily designed for typeset mathematics.

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1. Develop a LaTeX script to create a simple document that consists of 2 sections [Section1, Section2], and a paragraph with dummy text in each section. And also include header [title of document] and footer [institute name, page number] in the document.

Program:

```
\documentclass[12pt,a4paper]{article}
\usepackage[left=2cm,right=2cm,top=2cm,bottom=2cm]{geometry}
\usepackage{fancyhdr}
\begin{document}
% Set the page style to "fancy"...
\pagestyle{fancy}
\title{GNU Project}
\fancyhf{} % clear existing header/footer entries
% We don't need to specify the O coordinate
\fancyhead{} % clear all header fields
\fancyhead[R]{GNU Project}
\fancyfoot{}  % clear all footer fields
\fancyfoot[LO,CE]{Sri SaiRam College Of Engineering}
\fancyfoot[R]{\thepage}
\maketitle

\section{What is GNU?}
```

GNU is an operating system that is free software—that is, it respects users' freedom. The GNU operating system consists of GNU packages (programs specifically released by the GNU Project) as well as free software released by third parties. The development of GNU made it possible to use a computer without software that would trample your freedom.

```
\section{More about GNU}
```

GNU is a Unix-like operating system. That means it is a collection of many programs: applications, libraries, developer tools, even games. The development of GNU, started in January 1984, is known as the GNU Project. Many of the programs in GNU are released under the auspices of the GNU Project; those we call GNU packages. \\

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The name "GNU" is a recursive acronym for "GNU's Not Unix." "GNU" is pronounced g'noo, as one syllable, like saying "grew" but replacing the r with n. \\

The program in a Unix-like system that allocates machine resources and talks to the hardware is called the "kernel." GNU is typically used with a kernel called Linux. This combination is the GNU/Linux operating system. GNU/Linux is used by millions, though many call it "Linux" by mistake.

\section{What is the Free Software Movement?}
The free software movement campaigns to win for the users of computing the freedom that comes from free software. Free software puts its users in control of their own computing. Nonfree software puts its users under the power of the software's developer. \\

\section{What is Free Software?}
\textbf{Free software means the users have the freedom to run, copy, distribute, study, change and improve the software.}

Free software is a matter of liberty, not price. To understand the concept, you should think of "free" as in "free speech," not as in "free beer". More precisely, free software means users of a program have the four essential freedoms:

\begin{itemize}
\item The freedom to run the program as you wish, for any purpose (freedom 0).
\item The freedom to study how the program works, and change it so it does your computing as you wish (freedom 1). Access to the source code is a precondition for this.
\item The freedom to redistribute copies so you can help others (freedom 2).
\item The freedom to distribute copies of your modified versions to others (freedom 3). By doing this you can give the whole community a chance to benefit from your changes. Access to the source code is a precondition for this.

Developments in technology and network use have made these freedoms even more important now than they were in 1983. Nowadays the free software movement goes far beyond developing the GNU system.

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\end{itemize}

\end{document}

OUTPUT:

GNU Project

April 24, 2024

1 What is GNU?

GNU is an operating system that is free software—that is, it respects users' freedom. The GNU operating system consists of GNU packages (programs specifically released by the GNU Project) as well as free software released by third parties. The development of GNU made it possible to use a computer without software that would trample your freedom.

2 More about GNU

GNU is a Unix-like operating system. That means it is a collection of many programs: applications, libraries, developer tools, even games. The development of GNU, started in January 1984, is known as the GNU Project. Many of the programs in GNU are released under the auspices of the GNU Project; those we call GNU packages.

The name "GNU" is a recursive acronym for "GNU's Not Unix." "GNU" is pronounced g'noo, as one syllable, like saying "grew" but replacing the r with n.

The program in a Unix-like system that allocates machine resources and talks to the hardware is called the "kernel." GNU is typically used with a kernel called Linux. This combination is the GNU/Linux operating system. GNU/Linux is used by millions, though many call it "Linux" by mistake.

3 What is the Free Software Movement?

The free software movement campaigns to win for the users of computing the freedom that comes from free software. Free software puts its users in control of their own computing. Non-free software puts its users under the power of the software's developer.

4 What is Free Software?

Free software means the users have the freedom to run, copy, distribute, study, change and improve the software.

Free software is a matter of liberty, not price. To understand the concept, you should think of "free" as in "free speech," not as in "free beer". More precisely, free software means users of a program have the four essential freedoms:

- The freedom to run the program as you wish, for any purpose (freedom 0).

1

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- The freedom to study how the program works, and change it so it does your computing as you wish (freedom 1). Access to the source code is a precondition for this.
- The freedom to redistribute copies so you can help others (freedom 2).
- The freedom to distribute copies of your modified versions to others (freedom 3). By doing this you can give the whole community a chance to benefit from your changes. Access to the source code is a precondition for this.

  Developments in technology and network use have made these freedoms even more important now than they were in 1983. Nowadays the free software movement goes far beyond developing the GNU system.

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2. Develop a LaTeX script to create a document that displays the sample Abstract/Summary.

Program:

```
\documentclass[10pt,a4paper]{article}
\usepackage[utf8]{inputenc}
\usepackage{amsmath}
\usepackage{amsfonts}
\usepackage{amssymb}
\usepackage[left=3cm,right=3cm,top=2cm,bottom=2cm]{geometry}
%\usepackage{lipsum}
\begin{document}
\thispagestyle{plain}
\begin{center}
 \Large
 \textbf{Thesis Title}
 \vspace{0.4cm}
 \large
 Thesis Subtitle
 \vspace{0.4cm}
 \textbf{Author Name}
 \vspace{0.9cm}
 \textbf{Abstract}
\end{center}
%\lipsum[1]
```

News has become an important medium for everyone to stay aware and updated with the latest happenings in the world. It is important to note that in recent times with the rise in social media the amount of fake news has surged to dangerous levels. Fake news causes a lot of economic and social problems. It also gets difficult to trace back the source of fake news and hold someone accountable in order to curb it down. It is important to have a mechanism which evicts out fake news and contains only authentic news.

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Fake news causes a lot of economic and social problems. It also gets difficult to trace back the source of fake news and hold someone accountable in order to curb it down. It is important to have a mechanism whichevicts out fake news and contains only authentic news. With digitization, there has been a drastic increase in the usage of some of the popular social media sites such as Twitter, Facebook, Yahoo, YouTube as a medium of spreading news. There is very little check on the spreading of fake news. Accountability, in terms of how authentic the news is, is very less.

Our project aims to create a platform that runs on a blockchain network. News based media will be shared across this network. The origin of data and every user interaction will be recorded and get updated in the decentralized ledger and since it is in the blockchain, it remains immutable and nearly immune to any cyber threat.

The platform will be highly intuitive and interactive for users to browse through different sections of news. Users can view articles, blogs, headlines and other news-based content. Users can also push their own content on the network. The platform also allows the users to manually rate a piece of news based on its authenticity.
\end{document}

OUTPUT

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Thesis Title

Thesis Subtitle

Author Name

Abstract

News has become an important medium for everyone to stay aware and updated with the latest happenings in the world. It is important to note that in recent times with the rise in social media the amount of fake news has surged to dangerous levels. Fake news causes a lot of economic and social problems. It also gets difficult to trace back the source of fake news and hold someone accountable in order to curb it down. It is important to have a mechanism which evicts out fake news and contains only authentic news.

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Our project aims to create a platform that runs on a blockchain network. News based media will be shared across this network. The origin of data and every user interaction will be recorded and get updated in the decentralized ledger and since it is in the blockchain, it remains immutable and nearly immune to any cyber threat.

The platform will be highly intuitive and interactive for users to browse through different sections of news. Users can view articles, blogs, headlines and other news-based content. Users can also push their own content on the network. The platform also allows the users to manually rate a piece of news based on its authenticity.

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3. Develop a LaTeX script to create a simple title page of the VTU project Report [Use suitable Logos and text formatting]

Program:

Title.tex

```
\begin{titlingpage}
\thispagestyle{empty}
\centering
\setlength{\toptafiddle}{1in}
\setlength{\bottafiddle}{1in}
\vspace*{-0.75in}
\enlargethispage{\toptafiddle}
\begin{LARGE}
\textbf{VISVESVARAYA TECHNOLOGICAL UNIVERSITY}\\
\end{LARGE}
\begin{small}
(Belagavi-590018, Karnataka)
\end{small}
\begin{figure}[h]
   \centering
   \includegraphics[height=4cm]{vtu.png}
\end{figure}
\vfill
\Large{\textbf{\textcolor{therablue}{A PROJECT REPORT \\ ON\\ Employee Emotion
 Detection}}}\\
\Large{\textbf{Keeping Your Employees Happy}}
\vfill

\begin{small}
\textit{Submitted in the partial fulfillment of the requirements for the award of degree ,} \\
\end{small}
\begin{normalsize}
\textbf{\textit{Bachelor of Engineering }} \\
in \\
```

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```
\textbf{\textit{Computer Science and Engineering}} \\
by \\
\end{normalsize}
\vfill
\begin{tabular}{ccc}
\textbf{1. Student Name }&  & \textbf{1SB22CS001}\\
\textbf{2. Student Name }&  & \textbf{1SB22CS002}\\
\textbf{3. Student Name }&  & \textbf{1SB22CS003}\\
\textbf{4. Student Name }&  & \textbf{1SB22CS004}\\
\end{tabular}
\vfill
{\normalsize under the guidance of}\\
\Large{\textbf{Prof. ABC}}\\
Assistant Professor\\
\vfill
\begin{figure}[h]
   \centering
   \includegraphics[height=3.7cm]{sit.png}
\end{figure}
\textbf{Department of Computer Science \& Engineering \\ SRI SAIRAM COLLEGE \&
 ENGINEERING \\ ANEKAL, BENGALURU-562106 \\ ACADEMIC YEAR -2023-24}\\
\end{titlingpage}
```

FrontPage.tex

```
% Memoir is more versatile than other document classes like article, report, book etc,
\documentclass[11pt,a4paper,oneside]{memoir}
% To handle image, like scaling
\usepackage{graphicx}
\usepackage[english]{babel}
% Used for modifying page layout
\usepackage[a4paper,right=1in]{geometry}
% For referencing contents,figure,table...
\usepackage[hidelinks]{hyperref}
```


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```
% For including source code
\usepackage{listings}
% For using text colors
\usepackage{color}
% To attach pdf files
\usepackage{pdfpages}
% Document starts here
\begin{document}
\newlength{\toptafiddle}
\newlength{\bottafiddle}
\definecolor{therablue}{RGB}{41, 134, 230}

\include{Title} % Include Title Page, (i.e. title.tex)
% Page layout according to VTU specification
% Right: 1.25in, Left: 1in, Top & Bottom: 0.75in in each page
\setlength{\oddsidemargin}{0.25in} % Left side margin {1in by default + 0.25in}
% Header specification
\setlength{\headheight}{\onelineskip}
\setlength{\headsep}{6pt}
\setlength{\topmargin}{-0.25in}
% Footer specification
\setlength{\footskip}{\onelineskip}
\setlength{\footnotesep}{\onelineskip}
% A4 paper height = 11.69in
% thus 11.69in - 9.67in - 1in (top + header) is approx 0.75in left for bottom
\setlength{\textheight}{9.67in}
\brokenpenalty=10000 % Disallow page breaks at hyphens
\OnehalfSpacing % Line Spacing set to 1.5
%\pagenumbering{roman}
\pagestyle{plain}
\end{document}
```

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OUTPUT:

VISVESVARAYA TECHNOLOGICAL UNIVERSITY
(Belagavi-590018, Karnataka)



A PROJECT REPORT
ON
Employee Emotion Detection
Keeping Your Employees Happy

Submitted in the partial fulfillment of the requirements for the award of degree ,
Bachelor of Engineering
in
Computer Science and Engineering
by

1. Student Name 1SB22CS001
2. Student Name 1SB22CS002
3. Student Name 1SB22CS003
4. Student Name 1SB22CS004

under the guidance of
Prof. ABC
Assistant Professor



Sri SAIRAM COLLEGE OF ENGINEERING
Anekal, Bengaluru

Accredited by NAAC
ISO 9001:2015 Certified Institution
Approved by AICTE, New Delhi
Affiliated to Visvesvaraya Technological University
www.sairamce.edu.in

Department of Computer Science & Engineering
SRI SAIRAM COLLEGE & ENGINEERING
ANEKAL, BENGALURU-562106
ACADEMIC YEAR -2023-24

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4. Develop a LaTeX script to create the Certificate Page of the Report [Use suitable commands to leave the blank spaces for user entry]

Program:

Certificate.tex

```
\thispagestyle{empty}
\setlength{\toptafiddle}{1in}
\setlength{\bottafiddle}{1in}
\vspace*{-0.75in}
\enlargethispage{\toptafiddle}
\begin{center}
\begin{Large}
\textbf{Department of Computer Science and Engineering} \\
\end{Large}
\begin{Large}
\textbf{Sri Sairam College of Engineering}\\
\end{Large}
%\hspace{0.1cm}\\
\begin{small}
(Visvesvaraya Technological University, Belagavi,\\ Approved by AICTE, New Delhi, Accredited by NAAC and NBA and ISO 9001:2015 certified)
\end{small}
\vspace{0.2cm}
\begin{figure}[h]
\centering
\includegraphics[height=4cm]{vtu.png}
\hspace{0.1\textwidth}

\includegraphics[height=3.7cm]{sit.png}
\end{figure}
\Huge{Certificate}
\end{center}
\begin{large}
```

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```
This is to certify that the Project Report entitled
\textbf{"My Wonderful Project"}
is a bonafide work carried out by
\textbf{Student-1(1SB22CS001)},
\textbf{Student-2(1SB22CS002)},
\textbf{Student-3(1SB22CS003)}
and
\textbf{Student-4(1SI23CS004)}
```

in the partial fulfillment of the requirement for the award of the degree of Bachelor of Engineering in Computer Science and Engineering, Visvesvaraya Technological University, Belagavi during the year 2023-24. It is certified that all corrections/suggestions indicated for the internal assessment have been incorporated in the report. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the Bachelor of Engineering Degree.

```
\end{large}
\vfill
\vfill
\vfill
\begin{table}[h!]
\centering
\begin{tabular}{ccccccccc}
.....&&&&&&&&& .....\\
\textbf{{\footnotesize Guide}} &&&&&&&&&\textbf{{\footnotesize Group Convener}}\\
\textbf{Prof. XYZ}&&&&&&&&& \textbf{Dr. ABC} \\
\textbf{{\footnotesize Asst. Professor}} &&&&&&&&& \textbf{{\footnotesize Professor}}\\
\textbf{{\footnotesize Dept of CSE, SSCE}} &&&&&&&&& \textbf{{\footnotesize Dept of CSE, SSCE}}\\
\\
\\
.....&&&&&&&&& .....\\
\textbf{Dr. XXX} &&&&&&&&&  \textbf{Dr. YYY} \\
\textbf{{\footnotesize Professor and Head}} &&&&&&&&&  \textbf{{\footnotesize
  Principal}} \\
```


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```
\textbf{{\footnotesize Dept of CSE, SSCE}} &&&&&&&&&  \textbf{{\footnotesize SSCE,
 ANEKAL}}\\
\end{tabular}
\end{table}
Name of the Examiners          \hfill     Signature with Date
\begin{small}
\begin{enumerate}
\item Prof.
\item Prof.
\end{enumerate}
\end{small}
```

\FrontPage1.tex

```
% Memoir is more versatile than other document classes like article, report, book etc,
\documentclass[11pt,a4paper,oneside]{memoir}

% To handle image, like scaling
\usepackage{graphicx}
\usepackage[english]{babel}

% Used for modifying page layout
\usepackage[a4paper,right=1in]{geometry}

% For referencing contents,figure,table...
\usepackage[hidelinks]{hyperref}

% For including source code
\usepackage{listings}

% For using text colors
\usepackage{color}
```

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```
% To attach pdf files
\usepackage{pdfpages}

% Document starts here
\begin{document}

\newlength{\toptafiddle}
\newlength{\bottafiddle}
\definecolor{therablue}{RGB}{41, 134, 230}
%\include{Title} % Include Title Page, (i.e. title.tex)
% Page layout according to VTU specification
% Right: 1.25in, Left: 1in, Top & Bottom: 0.75in in each page
\setlength{\oddsidemargin}{0.25in} % Left side margin {1in by default + 0.25in}

% Header specification
\setlength{\headheight}{\onelineskip}
\setlength{\headsep}{6pt}
\setlength{\topmargin}{-0.25in}

% Footer specification
\setlength{\footskip}{\onelineskip}
\setlength{\footnotesep}{\onelineskip}

% A4 paper height = 11.69in
% thus 11.69in - 9.67in - 1in (top + header) is approx 0.75in left for bottom
\setlength{\textheight}{9.67in}
\brokenpenalty=10000 % Disallow page breaks at hyphens
\OnehalfSpacing % Line Spacing set to 1.5
%\pagenumbering{roman}
\include{Certificate}
\pagestyle{plain}
\end{document}
```

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OUTPUT:

Department of Computer Science and Engineering
Sri Sairam College of Engineering
(Visvesvaraya Technological University, Belagavi,
Approved by AICTE, New Delhi, Accredited by NAAC and NBA and ISO 9001:2015 certified)



Sri SAIRAM COLLEGE OF ENGINEERING
Anekal Bengaluru

Accredited by NAAC
ISO 9001:2015 Certified Institution
Approved by AICTE, New Delhi
Affiliated to Visvesvaraya Technological University
www.sairamce.edu.in

Certificate

This is to certify that the Project Report entitled **"My Wonderful Project"** is a bonafide work carried out by **Student-1(1SB22CS001)**, **Student-2(1SB22CS002)**, **Student-3(1SB22CS003)** and **Student-4(1SI23CS004)** in the partial fulfillment of the requirement for the award of the degree of Bachelor of Engineering in Computer Science and Engineering, Visvesvaraya Technological University, Belagavi during the year 2023-24. It is certified that all corrections/suggestions indicated for the internal assessment have been incorporated in the report.The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the Bachelor of Engineering Degree.

Guide **Prof. XYZ** Asst. Professor Dept of CSE, SSCE	Group Convener **Dr. ABC** Professor Dept of CSE, SSCE
Dr. XXX Professor and Head Dept of CSE, SSCE	**Dr. YYY** Principal SSCE, ANEKAL

2

Name of the Examiners	Signature with Date
1. Prof.	
2. Prof.	

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5. Develop a LaTeX script to create a document that contains the following table with proper labels.

S.No	USN	Student Name	Marks		
			Subject1	Subject2	Subject3
1	4XX22XX001	Name 1	89	60	90
2	4XX22XX002	Name 2	78	45	98
3	4XX22XX003	Name 3	67	55	59

Program:

```
\documentclass[10pt,a4paper]{article}
\usepackage[utf8]{inputenc}
\usepackage{amsmath}
\usepackage{amsfonts}
\usepackage{amssymb}
\usepackage[left=2cm, right=2cm,top=2cm,bottom=2cm]{geometry}
\usepackage{multirow}
\begin{document}
\begin{center}
\begin{Large}
\textbf{Table Demo}
\end{Large}
\end{center}
\section*{Marks Table}
\begin{tabular}{|c|c|c|c|c|c|}
\hline
\multirow{2}{*}{S.No} & \multirow{2}{*}{USN} & \multirow{2}{*}{Student Name} & \multicolumn{3}{c|}{Marks} \\
\cline{4-6}
 &  &  & Subject1 & Subject2 & Subject3 \\
\hline
1 & 4XX22XX001 & Name 1 & 88 & 77 & 97 \\
\hline
```


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```
2 & 4XX22XX002 & Name 2 & 74 & 78 & 66 \\
\hline
3 & 4XX22XX003 & Name 3 & 88 & 82 & 79 \\
\hline
\end{tabular}
\end{document}
```

OUTPUT

Table Demo

Marks Table

S.No	USN	Student Name	Marks		
			Subject1	Subject2	Subject3
1	4XX22XX001	Name 1	88	77	97
2	4XX22XX002	Name 2	74	78	66
3	4XX22XX003	Name 3	88	82	79

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6. Develop a LaTeX script to include the side-by-side graphics/pictures/figures in the document by using the subgraph concept.

Program:

```
\documentclass[10pt,a4paper]{article}
\usepackage[utf8]{inputenc}
\usepackage{amsmath}
\usepackage{amsfonts}
\usepackage{amssymb}
\usepackage{caption}
\usepackage{subcaption}
\usepackage{graphicx}
\usepackage[left=2cm,right=2cm,top=2cm,bottom=2cm]{geometry}
\begin{document}
\section*{Subfigure Demo}
\begin{figure}[h]
    \centering
    \begin{subfigure}[b]{0.3\textwidth}
        \centering
        \includegraphics[width=\textwidth]{graph1.png}
        \caption{$y=x$}
        \label{fig:y equals x}
    \end{subfigure}
    \hfill
    \begin{subfigure}[b]{0.3\textwidth}
        \centering
        \includegraphics[width=\textwidth]{graph2.png}
        \caption{$y=3\sin x$}
        \label{fig:three sin x}
    \end{subfigure}
    \hfill
    \begin{subfigure}[b]{0.3\textwidth}
        \centering
        \includegraphics[width=\textwidth]{graph3.png}
```

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```
    \caption{$y=5/x$}
    \label{fig:five over x}
  \end{subfigure}
  \caption{Three simple graphs arranged side-by-side}
  \label{fig:three graphs}
\end{figure}
\end{document}
```

OUTPUT:

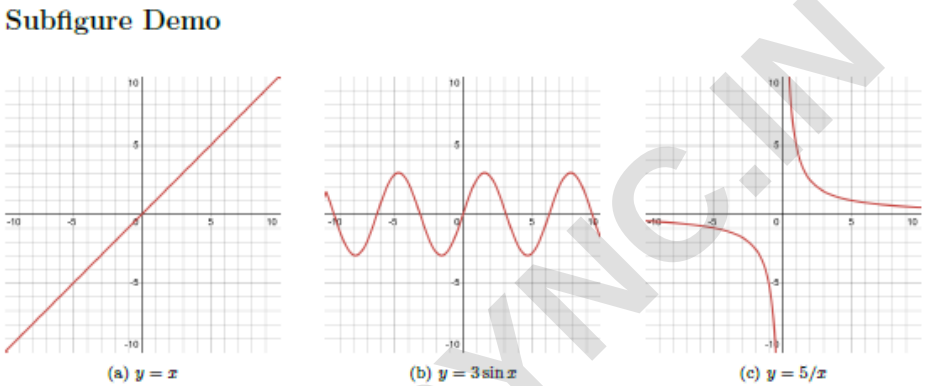


Figure 1: Three simple graphs arranged side-by-side

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7. Develop a LaTeX script to create a document that consists of the following two mathematical equations

$$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

$$= \frac{-2 \pm \sqrt{2^2-4*(1)*(-8)}}{2*1}$$

$$= \frac{-2 \pm \sqrt{4+32}}{2}$$

$$\varphi_\sigma^\lambda A_t = \sum_{\pi \in C_t} \operatorname{sgn}(\pi) \varphi_\sigma^\lambda \varphi_\pi^\lambda$$

$$= \sum_{\tau \in C_{\sigma t}} \operatorname{sgn}(\sigma^{-1}\tau\sigma) \varphi_\sigma^\lambda \varphi_{\sigma^{-1}\tau\sigma}^\lambda$$

$$= A_{\sigma t} \varphi_\sigma^\lambda$$

Program:

```
\documentclass[10pt,a4paper]{article}
\usepackage[utf8]{inputenc}
\usepackage{amsmath,nccmath}
\usepackage{amsfonts}
\usepackage{amssymb}
\usepackage[left=2cm,right=2cm,top=2cm,bottom=2cm]{geometry}
\begin{document}
\begin{center}
\Large{\textbf{Equations in \LaTeX}}
\end{center}
\section*{Equation 1}
%\begin{eqnarray}
%x = \frac{-b \pm \sqrt{b^{2}-4ac}}{2a} \\
%= \frac{-2 \pm \sqrt{2^{2}-4*(1)*(-8)}}{2*1}
%\end{eqnarray}
\begin{fleqn}
\[
x = \frac{-b \pm \sqrt{b^{2}-4ac}}{2a}
\]
\[
= \frac{-2 \pm \sqrt{2^{2}-4*(1)*(-8)}}{2*1}
\]
```

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```
\[
= \frac{-2 \pm \sqrt{4+32}}{2}
\]
\end{fleqn}
\section*{Equation 2}
\begin{fleqn}
\[
\varphi^{\lambda}_{\sigma}A_{t} = \sum_{\pi \in C_{t}} sgn(\pi)\varphi^{\lambda}_{\sigma}\varphi^{\lambda}_{\pi}
\]
\[
= \sum_{\tau \in C_{\sigma t}} sgn(\sigma^{-1}\tau\sigma)\varphi^{\lambda}_{\sigma}\varphi^{\lambda}_{\sigma^{-1}\tau\sigma}
\]
\[
= A_{\sigma t} \varphi^{\lambda}_{\sigma}
\]
\end{fleqn}
\end{document}
```

OUTPUT:

Equations in LaTeX

Equation 1

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-2 \pm \sqrt{2^2 - 4 * (1) * (-8)}}{2 * 1}$$

$$= \frac{-2 \pm \sqrt{4 + 32}}{2}$$

Equation 2

$$\varphi^{\lambda}_{\sigma}A_t = \sum_{\pi \in C_t} sgn(\pi)\varphi^{\lambda}_{\sigma}\varphi^{\lambda}_{\pi}$$

$$= \sum_{\tau \in C_{\sigma t}} sgn(\sigma^{-1}\tau\sigma)\varphi^{\lambda}_{\sigma}\varphi^{\lambda}_{\sigma^{-1}\tau\sigma}$$

$$= A_{\sigma t}\varphi^{\lambda}_{\sigma}$$

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8. Develop a LaTeX script to demonstrate the presentation of Numbered theorems, definitions, corollaries, and lemmas in the document.

Program:

```
\documentclass{article}
\usepackage[english]{babel}
\usepackage{amsthm}
\newtheorem{theorem}{Theorem}[section]
\newtheorem{corollary}{Corollary}[theorem]
\newtheorem{lemma}[theorem]{Lemma}
\theoremstyle{definition}
\newtheorem{definition}{Definition}[section]
\begin{document}
\section{Numbered theorems, definitions, corollaries and lemmas}
Theorems can easily be defined:
\begin{theorem}
Let \(f\) be a function whose derivative exists in every point, then \(f\) is
a continuous function.
\end{theorem}
\begin{theorem}[Pythagorean theorem]
\label{pythagorean}
This is a theorem about right triangles and can be summarised in the next
equation
\[ x^2 + y^2 = z^2 \]
\end{theorem}
And a consequence of theorem \ref{pythagorean} is the statement in the next
corollary.
\begin{corollary}
There's no right rectangle whose sides measure 3cm, 4cm, and 6cm.
\end{corollary}
You can reference theorems such as \ref{pythagorean} when a label is assigned.
\begin{lemma}
Given two line segments whose lengths are \(a\) and \(b\) respectively there is a
real number \(r\) such that \(b=ra\).
```

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```
\end{lemma}
\begin{definition}[Absolute value function]
The absolute value function can be specified as a two-part definition as follows: \\
$
|x| =
\left\{
        \begin{array}{ll}
                x  & \mbox{if } x \geq 0 \\
                -x & \mbox{if } x < 0
        \end{array}
\right.
$
\end{definition}
\end{document}
```

OUTPUT:

1 Numbered theorems, definitions, corollaries and lemmas

Theorems can easily be defined:

Theorem 1.1. *Let f be a function whose derivative exists in every point, then f is a continuous function.*

Theorem 1.2 (Pythagorean theorem)**.** *This is a theorem about right triangles and can be summarised in the next equation*

$$x^2 + y^2 = z^2$$

And a consequence of theorem 1.2 is the statement in the next corollary.

Corollary 1.2.1. *There's no right rectangle whose sides measure 3cm, 4cm, and 6cm.*

You can reference theorems such as 1.2 when a label is assigned.

Lemma 1.3. *Given two line segments whose lengths are a and b respectively there is a real number r such that $b = ra$.*

Definition 1.1 (Absolute value function)**.** The absolute value function can be specified as a two-part definition as follows:

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

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9. Develop a LaTeX script to create a document that consists of two paragraphs with a minimum of 10 citations in it and display the reference in the section.

Program:

Main.tex

```
\documentclass{article}

\usepackage{graphicx} % Required for inserting images

\begin{document}

\begin{center}

   \Large{\textbf{References Demo}}

\end{center}

\section{Introduction}

%\section{Related Work}
```

For disaster management, uncertainty handling is the main key problem. But, in Joint Service deployment and Requests Allocation~(JSR) domain, research work mainly uses the approaches such as deterministic optimization \cite{hardtoshare, multicell, bandwidth}, Lyapunov optimization \cite{dataintensive}, stochastic optimization, replication of services to achieve high reliability, and forecasting of user requests using machine learning without considering uncertainty. In deterministic optimization \cite{edgeuav}, request demand is known before the run. However, in online optimization, time is divided into slots and performs optimization per slot basis, which does not consider uncertain demand. Even if we used any probability distribution to model demand, it does not provide the correct model/pattern to define the uncertain data \cite{edgeuncertainty}. Using a replication approach to achieve high availability also incurs extra resource cost \cite{robust}. Using the forecasting method also, we can not predict the impact of uncertain events on the requests, which may lead to under-provisioning/over-provisioning resources to process the required tasks \cite{rsome}.

```
\section{Experiment Setup and Performance Parameters}
```


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To demonstrate the efficiency of the proposed approaches, we will simulate the scenario for an urban site affected by any natural calamity \cite{oilindustry}. To implement optimization models, we will use the IBM Cplex Optimizer tool \cite{cplex}.

\bibliographystyle{IEEEtran}

\bibliography{ref}

\end{document}

ref.bib

@ARTICLE{oilindustry,

author={Ngoenriang, Napat and Turner, Stephen John and Niyato, Dusit and Supittayapornpong, Sucha},

journal={IEEE Internet of Things Journal},

title={Joint UAV-Placement and Data Delivery in Aerial Inspection under Uncertainties}, year={2021},

volume={},

number={},

pages={1-1},

doi={10.1109/JIOT.2021.3113713}}

@ARTICLE{uavservice,

author={Qu, Yuben and Dai, Haipeng and Wang, Haichao and Dong, Chao and Wu, Fan and Guo, Song and Wu, Qihui}, journal={IEEE Journal on Selected Areas in Communications},

title={Service Provisioning for UAV-Enabled Mobile Edge Computing},

year={2021},

volume={39},

number={11},

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pages={3287-3305},

doi={10.1109/JSAC.2021.3088660}

}

@misc{cplex,

author = {IBM},

title = {IBM CPLEX Optimizer},

howpublished = "\url{https://www.ibm.com/in-en/analytics/cplex-optimizer}",

year = {2021},

note = "[Online; accessed 3-Feb-2022]"

}

@misc{rsome,

author = {NSU},

title = {RSOME},

howpublished = "\url{https://xiongpengnus.github.io/rsome/}",

year = {2021},

note = "[Online; accessed 3-Feb-2022]"

}

@INPROCEEDINGS{hardtoshare,

author={He, Ting and Khamfroush, Hana and Wang, Shiqiang and La Porta, Tom and Stein, Sebastian},

booktitle={IEEE 38th International Conference on Distributed Computing Systems (ICDCS)},

title={It's Hard to Share: Joint Service Placement and Request Scheduling in Edge Clouds with Sharable and Non-Sharable Resources}, year={2018},

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```
  volume={},
  number={},
  pages={365-375},
  doi={10.1109/ICDCS.2018.00044}
}
@INPROCEEDINGS{multicell,
  author={Poularakis, Konstantinos and Llorca, Jaime and Tulino, Antonia M. and Taylor, Ian and Tassiulas, Leandros},
  booktitle={IEEE Conference on Computer Communications (INFOCOM)},
  title={Joint Service Placement and Request Routing in Multi-cell Mobile Edge Computing Networks},
  year={2019},
  volume={},
  number={},
  pages={10-18},
  doi={10.1109/INFOCOM.2019.8737385}
}
@ARTICLE{bandwidth,
  author={Poularakis, Konstantinos and Llorca, Jaime and Tulino, Antonia M. and Taylor, Ian},
  journal={IEEE/ACM Transactions on Networking},
  title={Service Placement and Request Routing in MEC Networks With Storage, Computation, and Communication Constraints},
  year={2020},
```

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volume={28},

number={3},

pages={1047-1060},

doi={10.1109/TNET.2020.2980175}

}

@INPROCEEDINGS{dataintensive,

author={Farhadi, Vajiheh and Mehmeti, Fidan and He, Ting and Porta, Tom La and Khamfroush, Hana and Wang, Shiqiang and Chan, Kevin S},

booktitle={IEEE Conference on Computer Communications(INFOCOM)},

title={Service Placement and Request Scheduling for Data-intensive Applications in Edge Clouds},

year={2019},

volume={},

number={},

pages={1279-1287},

doi={10.1109/INFOCOM.2019.8737368}

}

@ARTICLE{resource,

author={Ahmed, Shakil and Chowdhury, Mostafa Zaman and Sabuj, Saifur Rahman and Alam, Md Imtiajul and Jang, Yeong Min}, journal={IEEE Access}, title={Energy-Efficient UAV Relaying Robust Resource Allocation in Uncertain Adversarial Networks}, year={2021}, volume={9}, number={}, pages={59920-59934}, doi={10.1109/ACCESS.2021.3073015}}

@ARTICLE{resource2, author={Yang, Zhaohui and Pan, Cunhua and Wang, Kezhi and Shikh-Bahaei, Mohammad}, journal={IEEE Transactions on Wireless Communications}, title={Energy Efficient Resource Allocation in UAV-Enabled Mobile Edge Computing

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Networks}, year={2019}, volume={18}, number={9}, pages={4576-4589}, doi={10.1109/TWC.2019.2927313}}

@ARTICLE{offload, author={Apostolopoulos, Pavlos Athanasios and Fragkos, Georgios and Tsiropoulou, Eirini Eleni and Papavassiliou, Symeon}, journal={IEEE Transactions on Mobile Computing}, title={Data Offloading in UAV-assisted Multi-access Edge Computing Systems under Resource Uncertainty}, year={2021}, volume={}, number={}, pages={1-1}, doi={10.1109/TMC.2021.3069911}}

@INPROCEEDINGS{offload2, author={Zhou, Fuhui and Wu, Yongpeng and Sun, Haijian and Chu, Zheng}, booktitle={2018 IEEE International Conference on Communications (ICC)}, title={UAV-Enabled Mobile Edge Computing: Offloading Optimization and Trajectory Design}, year={2018}, volume={}, number={}, pages={1-6}, doi={10.1109/ICC.2018.8422277}}

@ARTICLE{trajectory, author={Wang, Kai and Zhang, Xiao and Duan, Lingjie and Tie, Jun}, journal={IEEE Transactions on Mobile Computing}, title={Multi-UAV Cooperative Trajectory for Servicing Dynamic Demands and Charging Battery}, year={2021}, volume={}, number={}, pages={1-1}, doi={10.1109/TMC.2021.3110299}}

@article{edgeuncertainty,

author = {Xu, Xiaolong and Cao, Hao and Geng, Qingfan and Liu, Xihua and Dai, Fei and Wang, Chuanjian},

title = {Dynamic resource provisioning for workflow scheduling under uncertainty in edge computing environment},

journal = {Concurrency and Computation: Practice and Experience},

volume = {n/a},

number = {n/a},

pages = {e5674},

keywords = {edge computing, SDN, uncertainty, workflow scheduling},

doi = {https://doi.org/10.1002/cpe.5674}

}

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@ARTICLE{edgeuav, author={Qu, Yuben and Dai, Haipeng and Wang, Haichao and Dong, Chao and Wu, Fan and Guo, Song and Wu, Qihui}, journal={IEEE Journal on Selected Areas in Communications}, title={Service Provisioning for UAV-Enabled Mobile Edge Computing}, year={2021}, volume={39}, number={11}, pages={3287-3305}, doi={10.1109/JSAC.2021.3088660}}

@inproceedings{mobility,

title={UAV 3D Mobility Model Oriented to Dynamic and Uncertain Environment},

author={Na Wang and Nan Di and Fei Dai and Fangxin Liu},

booktitle={ICA3PP},

year={2018}

}

@ARTICLE{robust, author={Li, Bo and He, Qiang and Cui, Guangming and Xia, Xiaoyu and Chen, Feifei and Jin, Hai and Yang, Yun}, journal={IEEE Transactions on Services Computing}, title={READ: Robustness-oriented Edge Application Deployment in Edge Computing Environment}, year={2020}, volume={}, number={}, pages={1-1}, doi={10.1109/TSC.2020.3015316}}

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OUTPUT:

References Demo

1 Introduction

For disaster management, uncertainty handling is the main key problem. But, in Joint Service deployment and Requests Allocation (JSR) domain, research work mainly uses the approaches such as deterministic optimization [1, 2, 3], Lyapunov optimization [4], stochastic optimization, replication of services to achieve high reliability, and forecasting of user requests using machine learning without considering uncertainty. In deterministic optimization [5], request demand is known before the run. However, in online optimization, time is divided into slots and performs optimization per slot basis, which does not consider uncertain demand. Even if we used any probability distribution to model demand, it does not provide the correct model/pattern to define the uncertain data [6]. Using a replication approach to achieve high availability also incurs extra resource cost [7]. Using the forecasting method also, we can not predict the impact of uncertain events on the requests, which may lead to under-provisioning/over-provisioning resources to process the required tasks [8].

2 Experiment Setup and Performance Parameters

To demonstrate the efficiency of the proposed approaches, we will simulate the scenario for an urban site affected by any natural calamity [9]. To implement optimization models, we will use the IBM Cplex Optimizer tool [10].

References

[1] T. He, H. Khamfroush, S. Wang, T. La Porta, and S. Stein, "It's hard to share: Joint service placement and request scheduling in edge clouds with sharable and non-sharable resources," in *IEEE 38th International Conference on Distributed Computing Systems (ICDCS)*, 2018, pp. 365–375.

[2] K. Poularakis, J. Llorca, A. M. Tulino, I. Taylor, and L. Tassiulas, "Joint service placement and request routing in multi-cell mobile edge computing networks," in *IEEE Conference on Computer Communications (INFOCOM)*, 2019, pp. 10–18.

[3] K. Poularakis, J. Llorca, A. M. Tulino, and I. Taylor, "Service placement and request routing in mec networks with storage, computation, and communication constraints," *IEEE/ACM Transactions on Networking*, vol. 28, no. 3, pp. 1047–1060, 2020.

[4] V. Farhadi, F. Mehmeti, T. He, T. L. Porta, H. Khamfroush, S. Wang, and K. S. Chan, "Service placement and request scheduling for data-intensive

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applications in edge clouds," in *IEEE Conference on Computer Communications(INFOCOM)*, 2019, pp. 1279–1287.

[5] Y. Qu, H. Dai, H. Wang, C. Dong, F. Wu, S. Guo, and Q. Wu, "Service provisioning for uav-enabled mobile edge computing," *IEEE Journal on Selected Areas in Communications*, vol. 39, no. 11, pp. 3287–3305, 2021.

[6] X. Xu, H. Cao, Q. Geng, X. Liu, F. Dai, and C. Wang, "Dynamic resource provisioning for workflow scheduling under uncertainty in edge computing environment," *Concurrency and Computation: Practice and Experience*, vol. n/a, no. n/a, p. e5674.

[7] B. Li, Q. He, G. Cui, X. Xia, F. Chen, H. Jin, and Y. Yang, "Read: Robustness-oriented edge application deployment in edge computing environment," *IEEE Transactions on Services Computing*, pp. 1–1, 2020.

[8] NSU, "Rsome," https://xiongpengnus.github.io/rsome/, 2021, [Online; accessed 3-Feb-2022].

[9] N. Ngoenriang, S. J. Turner, D. Niyato, and S. Supittayapornpong, "Joint uav-placement and data delivery in aerial inspection under uncertainties," *IEEE Internet of Things Journal*, pp. 1–1, 2021.

[10] IBM, "Ibm cplex optimizer," https://www.ibm.com/in-en/analytics/cplex-optimizer, 2021, [Online; accessed 3-Feb-2022].

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10. Develop a LaTeX script to design a simple tree diagram or hierarchical structure in the document with appropriate labels using the Tikz library.

Program:

```
\documentclass{article}
\usepackage{tikz}
\begin{document}
\begin{center}
\Large{\textbf{Hierarchy of Linux distributions}}
\end{center}
\begin{figure}[h]
   \centering
\begin{tikzpicture} [every node/.style = {shape=rectangle, rounded corners, draw, align=center}]
  \path [draw,thick,-]
    node (root)[red] {GNU/Linux}
    [sibling distance=45mm, level distance=25mm]
    child {node [cyan] {Debian}
    [sibling distance=25mm, level distance=25mm]
          child { node [cyan] {Ubuntu} }
          child { node [cyan] {Linux Mint} }
%         child { node {Elementary} }
    }
    child {node [magenta] {RedHat}
    [sibling distance=25mm, level distance=25mm]
          child { node [magenta] {Fedora} }
          child { node [magenta] {OpenSuse} }
    }
    child {node [blue] {Arch}
          [sibling distance=25mm, level distance=25mm]
                child { node [blue]{Manjaro} }
                child { node [blue]{EndeavourOS} }
    };
```

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```
\end{tikzpicture}
 \caption{GNU/Linux Operating System Family}
\end{figure}
\pagebreak
\begin{center}
\Large{\textbf{SUV Cars}}
\end{center}
\begin{figure}[h]
   \centering
\begin{tikzpicture} [every node/.style = {shape=rectangle, rounded corners, draw, align=center}]
  \path [draw,thick,-]
        [grow=-45]
    node (root)[red] {SUV}
    [sibling distance=45mm, level distance=25mm]

    child {node [cyan] {Tata}
    [sibling distance=25mm, level distance=25mm]
        child { node [cyan] {Nexon} }
        child { node [cyan] {Punch} }
%       child { node {Elementary} }
    }
    child {node [magenta] {Volkswagen}
    [sibling distance=25mm, level distance=25mm]
        child { node [magenta] {Taigun} }
        child { node [magenta] {Virtus} }
    }
    child {node [blue] {Maruti}
        [sibling distance=25mm, level distance=25mm]
            child { node [blue]{Brezza} }
            child { node [blue]{Vitara} }
    };
```


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```
\end{tikzpicture}
 \caption{Car Brands Hierarchy}
\end{figure}
\end{document}
```

OUTPUT:

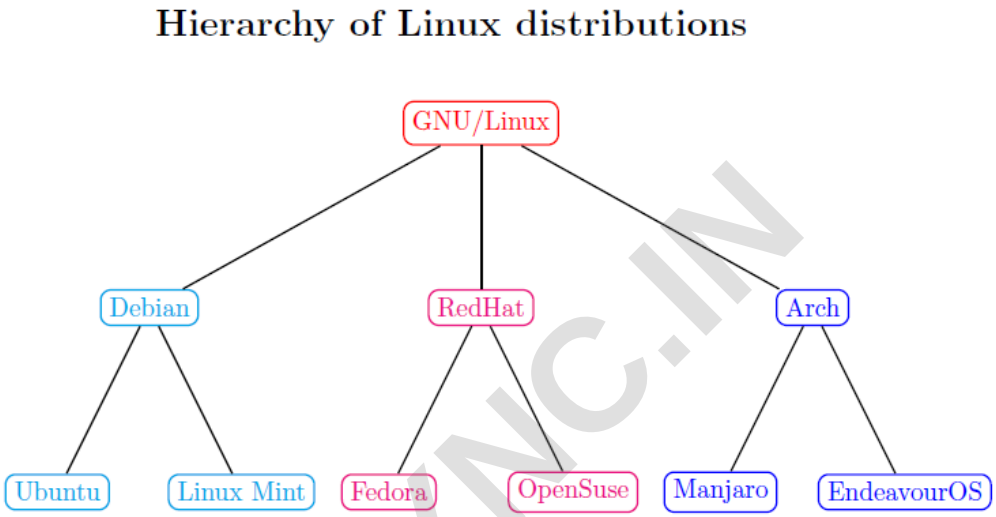


Figure 1: GNU/Linux Operating System Family

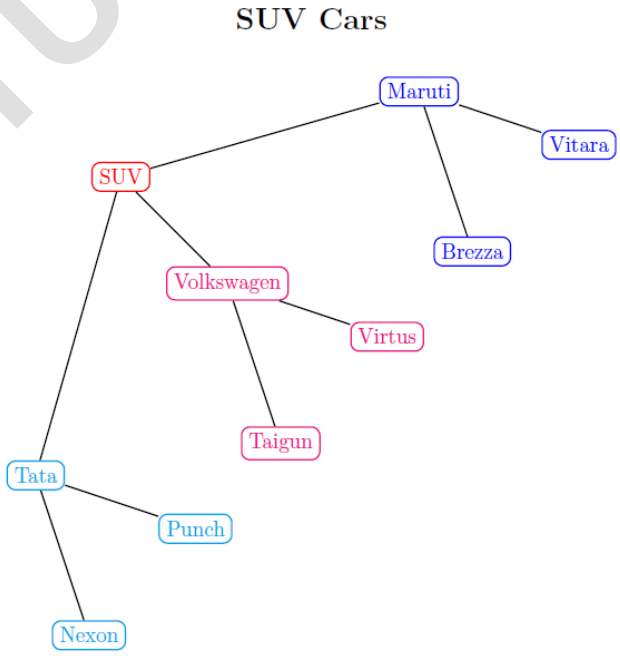


Figure 2: Car Brands Hierarchy

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11. Develop a LaTeX script to present an algorithm in the document using algorithm/algorithmic/algorithm2e library.

Program:

```
\documentclass[10pt,a4paper]{article}
\usepackage[utf8]{inputenc}
\usepackage{amsmath}
\usepackage{amsfonts}
\usepackage{amssymb}
\usepackage{algorithm2e}
\usepackage[left=2cm,right=2cm,top=2cm,bottom=2cm]{geometry}
\begin{document}
\section*{Floyd's Algorithm}
Algorithm to find solution to All-Pairs Shortest-Paths Problem
\SetKwComment{Comment}{// }{ }
\vspace{1cm}
        \begin{algorithm}[H]
      \caption{Floyd(W [1..n, 1..n])}
     \SetAlgoLined
      \DontPrintSemicolon
      \KwIn{The weight matrix W of a graph having vertices [1..n]}
      \KwOut{The distance matrix D of the shortest paths' lengths between every pair of vertices [1..n]}
      $D$ $\gets$ $W$ \Comment*[r]{initially copy the weight matrix into distance matrix}
      \For{$k \gets 1$ to $n$}{
        \For{$i \gets 1$ to $n$}{
            \For{$j \gets 1$ to $n$}{
                $D[i,j]$ $\gets min\lbrace D[i,j], D[i,k] + D[k,j]\rbrace$
            }
        }
      }
      \Return{$D$}\;
        \end{algorithm}
\end{document}
```


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OUTPUT:

Floyd's Algorithm

Algorithm to find solution to All-Pairs Shortest-Paths Problem

Input: The weight matrix W of a graph having vertices [1..n]
Output: The distance matrix D of the shortest paths' lengths between every pair of vertices [1..n]

```
D ← W                                  // initially copy the weight matrix into distance matrix
for k ← 1 to n do
    for i ← 1 to n do
        for j ← 1 to n do
            D[i, j] ← min{D[i, j], D[i, k] + D[k, j]}
        end
    end
end
return D
```

Algorithm 1: Floyd(W [1..n, 1..n])

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12. Develop a LaTeX script to create a simple report and article by using suitable commands and formats of user choice.

Program:

```
\documentclass[6pt,a4paper]{report}
\usepackage[utf8]{inputenc}
\usepackage{amsmath}
\usepackage{amsfonts}
\usepackage{amssymb}
\usepackage{graphicx}
\usepackage[left=3cm,right=3cm,top=2cm,bottom=2cm]{geometry}
\author{by XYZ}
\title{A Simple Report}
\begin{document}
\maketitle
\chapter{Free Software}
\section*{What is Free Software?}
"\textbf{Free software}" means software that respects users' freedom and community. Roughly, it means that \textbf{the users have the freedom to run, copy, distribute, study, change and improve the software}. Thus, "free software" is a matter of liberty, not price. To understand the concept, you should think of "\textit{free}" as in "\textit{free speech},"  not as in "\textit{free beer}." We sometimes call it "\textbf{libre software}," borrowing the French or Spanish word for "free" as in freedom, to show we do not mean the software is gratis.

You may have paid money to get copies of a free program, or you may have obtained copies at no charge. But regardless of how you got your copies, you always have the freedom to copy and change the software, even to sell copies.

We campaign for these freedoms because everyone deserves them. With these freedoms, the users (both individually and collectively) control the program and what it does for them. When users don't control the program, we call it a "\textit{nonfree}" or "\textit{proprietary}" program. The nonfree program controls the users, and the developer controls the program; this makes the program an instrument of unjust power.
```


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"\emph{Open source}" is something different: it has a very different philosophy based on different values. Its practical definition is different too, but nearly all open source programs are in fact free.

\section*{The Free Software Definition}

The free software definition presents the criteria for whether a particular software program qualifies as free software. \\

\textbf{The four essential freedoms} \\

A program is free software if the program's users have the four essential freedoms: \\

\begin{itemize}

\item The freedom to run the program as you wish, for any purpose (freedom 0).

\item The freedom to study how the program works, and change it so it does your computing as you wish (freedom 1). Access to the source code is a precondition for this.

\item The freedom to redistribute copies so you can help others (freedom 2).

\item The freedom to distribute copies of your modified versions to others (freedom 3).

\end{itemize}

By doing this you can give the whole community a chance to benefit from your changes. Access to the source code is a precondition for this. \\

A program is free software if it gives users adequately all of these freedoms. Otherwise, it is nonfree. While we can distinguish various nonfree distribution schemes in terms of how far they fall short of being free, we consider them all equally unethical.

\chapter{Listing Environment}

\begin{small}

\section*{Unordered lists}

\subsection*{Groceries List}

\begin{itemize}

\item Eggs

\item Milk

\item Biscuits

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```
        \item Rice
\end{itemize}
\subsection*{Football Teams}
\begin{itemize}
        \item English Premier League
        \begin{itemize}
                \item Manchester United
                \item Liverpool
        \end{itemize}

        \item La Liga
        \begin{itemize}
                \item Barcelona
                \item Real Madrid
        \end{itemize}

        \item Bundesliga
        \begin{itemize}
                \item Bayern Munich
                \item Borussia Dortmund
        \end{itemize}
\end{itemize}
\section*{Ordered lists}
\subsection*{ICC WTC Rankings}
\begin{enumerate}
\item India
\item Australia
\item New Zealand
\end{enumerate}
\subsection*{Countries ranked by Market Cap}
\begin{enumerate}
        \item Asia
        \begin{enumerate}
```

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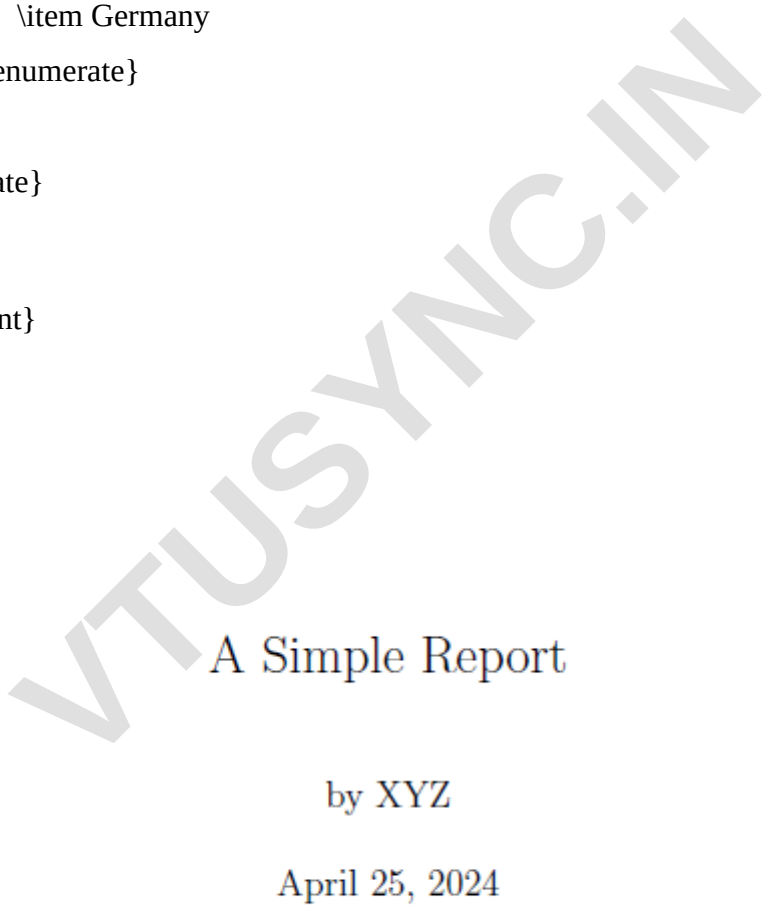
```
            \item China
            \item Japan
            \item India
        \end{enumerate}

        \item Europe
        \begin{enumerate}
            \item United Kingdom
            \item France
            \item Germany
        \end{enumerate}

\end{enumerate}
\end{small}

\end{document}
```

OUTPUT:



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Chapter 1

Free Software

What is Free Software?

"**Free software**" means software that respects users' freedom and community. Roughly, it means that **the users have the freedom to run, copy, distribute, study, change and improve the software**. Thus, "free software" is a matter of liberty, not price. To understand the concept, you should think of "*free*" as in "*free speech*," not as in "*free beer*." We sometimes call it "**libre software**," borrowing the French or Spanish word for "free" as in freedom, to show we do not mean the software is gratis.

You may have paid money to get copies of a free program, or you may have obtained copies at no charge. But regardless of how you got your copies, you always have the freedom to copy and change the software, even to sell copies.

We campaign for these freedoms because everyone deserves them. With these freedoms, the users (both individually and collectively) control the program and what it does for them. When users don't control the program, we call it a "*nonfree*" or "*proprietary*" program. The nonfree program controls the users, and the developer controls the program; this makes the program an instrument of unjust power.

"*Open source*" is something different: it has a very different philosophy based on different values. Its practical definition is different too, but nearly all open source programs are in fact free.

The Free Software Definition

The free software definition presents the criteria for whether a particular software program qualifies as free software.

The four essential freedoms

A program is free software if the program's users have the four essential freedoms:

- The freedom to run the program as you wish, for any purpose (freedom 0).
- The freedom to study how the program works, and change it so it does your computing as you wish (freedom 1). Access to the source code is a precondition for this.
- The freedom to redistribute copies so you can help others (freedom 2).
- The freedom to distribute copies of your modified versions to others (freedom 3).

By doing this you can give the whole community a chance to benefit from your changes. Access to the source code is a precondition for this.

A program is free software if it gives users adequately all of these freedoms. Otherwise, it is nonfree. While we can distinguish various nonfree distribution schemes in terms of how far they fall short of being free, we consider them all equally unethical.

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Chapter 2

Listing Environment

Unordered lists

Groceries List

- Eggs
- Milk
- Biscuits
- Rice

Football Teams

- English Premier League
  - Manchester United
  - Liverpool
- La Liga
  - Barcelona
  - Real Madrid
- Bundesliga
  - Bayern Munich
  - Borussia Dortmund

Ordered lists

ICC WTC Rankings

1. India
2. Australia
3. New Zealand

Countries ranked by Market Cap

1. Asia
   (a) China
   (b) Japan
   (c) India
2. Europe
   (a) United Kingdom
   (b) France
   (c) Germany

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Viva Question And Answers

1. Can you explain how LaTeX differs from other typesetting systems?

LaTeX is a high-quality typesetting system, distinct from others due to its focus on content over form. Unlike WYSIWYG systems like Word, where users control layout directly, LaTeX uses predefined templates allowing authors to concentrate on the text. It excels in creating documents with complex mathematical formulas, scientific and technical texts. Its programmability allows customization, automation of repetitive tasks, and integration into workflows. LaTeX's output quality surpasses other systems, producing professional-looking documents.

2. What would be the best way to import a large number of image files in LaTeX?

To import a large number of image files in LaTeX, the best approach would be to use loops with the \includegraphics command. This can be achieved by using the 'forloop' package or pgffor package's \foreach command.

For instance, if you have sequentially named images like img1.jpg, img2.jpg, etc., you could use:

```
\usepackage{pgffor}
…
\begin{document}
\foreach \i in {1,…,n} {\includegraphics[width=\linewidth]{img\i}}
\end{document}
```

Replace 'n' with the total number of images. The loop will automatically insert each image from img1 to imgn. Ensure that all images are in the same directory as your .tex file or specify the path correctly.

3. Can you describe briefly how to create tables using LaTeX?

To create tables in LaTeX, use the "tabular" environment. Start by declaring it with "\begin{tabular}{cols}", where 'cols' defines column alignment (l for left, c for center, r for

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right). Each row is separated by "\\" and columns within rows are divided by "&". For example:

```
\begin{tabular}{lcr}
Cell1 & Cell2 & Cell3 \\
Cell4 & Cell5 & Cell6 \\
\end{tabular}
```

Horizontal lines can be added using "\hline", vertical lines by adding "|" to the column definition. To merge cells horizontally, use "\multicolumn{num}{align}{content}". Merging vertically requires additional packages like multirow.

4. Explain how LaTeX handles reference citations by default and any potential issues that might arise.

LaTeX uses a two-pass system for reference citations. In the first pass, it records citation keys and line numbers in an auxiliary file while processing the document. During the second pass, it reads this file to resolve all references. This process can cause issues if the document is not compiled twice, leading to unresolved references.

Another potential issue arises from LaTeX's case-sensitivity with respect to citation keys. If there are discrepancies between the key used in the \cite command and the one defined in the bibliography, the reference will not be resolved correctly.

Additionally, LaTeX does not handle multiple bibliographies well by default. It requires additional packages like multibib or chapterbib to manage separate bibliographies for different sections of a document.

Lastly, LaTeX doesn't support dynamic referencing out-of-the-box. Packages such as hyperref are needed to create clickable links within the document.

5. How can you create a multi-column layout in LaTeX?

In LaTeX, a multi-column layout can be created using the "multicol" package. This is done by including \usepackage{multicol} in the preamble of your document. To start a multi-column section, use the command \begin{multicols}{n}, where 'n' represents the number of

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columns you want. The text that follows will automatically flow into the specified number of columns until you end the multi-column section with \end{multicols}. For example:

```
\documentclass{article}
\usepackage{multicol}
\begin{document}
\begin{multicols}{2}
Lorem ipsum dolor sit amet…
\end{multicols}
\end{document}
```

This creates a two-column layout for the text within the multicol environment.

6. Can you explain the difference between “\newline” and “\\” in LaTeX?

In LaTeX, both “\newline” and “\\” are used to create line breaks. However, they function differently. “\\ ” is a more flexible command that allows for optional arguments such as extra vertical space or alignment in tabular environments. It can also be used multiple times in succession to create blank lines. On the other hand, “\newline” is stricter; it doesn’t accept any argument and using it twice in a row will result in an error. Therefore, while both commands serve similar purposes, their usage depends on the specific requirements of your document’s formatting.

7. Can you explain how to control white spaces in LaTeX?

In LaTeX, controlling white spaces is achieved through several commands. The “\ ” command inserts a non-breaking space, preventing line breaks at its location. For larger spaces, use the “\quad” and “\qquad” commands for 1em and 2em spaces respectively.

To create vertical spaces, use the “\vspace{size}” command where size can be in pt, mm, cm, etc. To ensure this space even at the beginning or end of a page, use “\vspace*{size}”.

For horizontal spacing, “\hspace{size}” is used similarly to “\vspace”. Use “\hspace*” to force the space at the start or end of a line.

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The “\newline” command starts a new line without starting a new paragraph. The “\par” command starts a new paragraph with an indent unless changed by “\noindent”.

8. How would you create a LaTeX command that takes an argument?

To create a LaTeX command that takes an argument, you use the \newcommand macro. The syntax is \newcommand{\commandname}[number of arguments]{definition}. For instance, to define a command named “\hello” that takes one argument, we write:

\newcommand{\hello}[1]{Hello, #1}

Here, “#1” represents the first argument passed to the command. When using this new command in your document, you would write it as \hello{World}, which will output “Hello, World”.

9. How can you modify the layout of a LaTeX-generated PDF to customize it to your needs?

LaTeX provides several packages to modify the layout of a PDF. The ‘geometry’ package allows you to adjust margins, paper size and orientation. Use \usepackage{geometry} in your preamble, then specify parameters like \geometry{a4paper, left=20mm, right=20mm, top=25mm, bottom=30mm}.

The ‘fancyhdr’ package customizes headers and footers. Include it with \usepackage{fancyhdr}, then define your header/footer using \lhead{}, \chead{}, \rhead{} for left, center, right header respectively, and similarly for footer.

For altering text appearance, use commands like \textbf{} for bold, \textit{} for italic, or \underline{} for underline. For font size, use commands from \tiny to \Huge.

To customize sections, use \section{}, \subsection{}, \subsubsection{}. To alter their formatting, redefine them using \renewcommand{\thesection}{…} where … is your new format.

10. Can you explain the difference between bibtex and biblatex?

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BibTeX and BibLaTeX are both bibliography management systems in LaTeX, but they differ significantly. BibTeX is the traditional system, which formats bibliographies according to predefined styles such as IEEE or APA. It's less flexible than biblatex, but it's widely supported by journal publishers.

On the other hand, BibLaTeX is a modern package that provides more control over citation styles. It uses Biber as its default backend, which supports Unicode, complex sorting mechanisms, and multiple bibliographies. However, it may not be compatible with some publisher templates.

11. How would you create a custom document class in LaTeX?

To create a custom document class in LaTeX, you need to write a .cls file. This file defines the new class and its behavior. Start by copying an existing class file like article.cls as your base. Rename it to yourclassname.cls. Open this file in a text editor and replace all instances of the original class name with your new class name. Modify the commands and environments according to your needs. You can define new commands using \newcommand or modify existing ones using \renewcommand. Similarly, use \newenvironment and \renewenvironment for environments. Save the changes and place the .cls file in the directory where you'll be working on your .tex files. To use your new class, start your .tex file with \documentclass{yourclassname}.

12. Describe how to write mathematical symbols and equations in LaTeX.

In LaTeX, mathematical symbols and equations are written within math environments. Inline math is denoted by dollar signs ($), while display math uses double dollar signs ($$). For example, $a^2 + b^2 = c^2$.

Symbols are represented with backslashes followed by the symbol's name. For instance, Greek letters like α, β, or π are written as \alpha, \beta, \pi respectively. Common operators such as plus (+), minus (-), times (*) and divide (/) can be used directly.

For more complex equations, use the equation environment: \begin{equation}…\end{equation}. This allows for automatic numbering. Unnumbered equations use the asterisk variant: \begin{equation*}…\end{equation*}.

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Fractions are created using \frac{numerator}{denominator}, square roots with \sqrt{radicand}, and sums with \sum_{lower limit}^{upper limit}. Subscripts and superscripts are indicated by underscore (_) and caret (^) respectively.

13. What are some commands to adjust the margins and padding of the text?

In LaTeX, the 'geometry' package is used to adjust margins and padding. The command '\usepackage[margin=1in]{geometry}' sets all margins to 1 inch. For individual adjustments, use '\usepackage[left=2cm,right=3cm,top=2cm,bottom=3cm]{geometry}'. Padding can be adjusted using the '\setlength{\parindent}{0pt}' command for no indentation or '\setlength{\parskip}{10pt plus 1pt minus 1pt}' for space between paragraphs

14. In LaTeX, how would you handle special characters, such as #, $, or %?

In LaTeX, special characters are handled by prefixing them with a backslash. For instance, to insert the hash symbol (#), you would write \#. Similarly, for dollar ($) and percentage (%) symbols, you would use \$ and \% respectively. However, note that some special characters like the backslash itself require different handling. To print a single backslash, you need to use two backslashes (\\). This is because in LaTeX, the backslash is used as an escape character, signaling that the next character has a special meaning.

15. How would you define new commands or environments in LaTeX?

In LaTeX, new commands are defined using the \newcommand command. The syntax is: \newcommand{\commandname}[number of arguments][default value for first argument]{definition}. For example, to define a command that bolds and italicizes text, you could write: \newcommand{\bolditalic}[1]{\textbf{\textit{#1}}}.

Environments are defined with \newenvironment. Its syntax is: \newenvironment{name}[num]{before}{after}. 'Before' code executes at the beginning of the environment and 'after' code at its end. For instance, to create an indented quote environment, use: \newenvironment{indentquote}{\begin{quote}\begin{adjustwidth}{2em}{2em}}{\end{adjustwidth}\end{quote}}.

16. Can you describe how to create a custom bibliography style in LaTeX?

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To create a custom bibliography style in LaTeX, you need to use the BibTeX tool. First, generate a .bst file using the command “latex makebst.tex”. This will initiate an interactive session where you can specify your preferences for the bibliography style. Once completed, it generates a .bst file that defines your custom style.

Next, include this style in your document by adding “\bibliographystyle{yourstyle}” before “\begin{document}”. Replace ‘yourstyle’ with the name of your .bst file (without extension).

Then, create a .bib file containing your references. Each entry should have a unique key and fields like author, title, year, etc. Use “@article”, “@book” etc., at the start of each entry to define its type.

Finally, cite these entries in your document using “\cite{key}”, where ‘key’ is the unique identifier for each reference. Compile your document using pdflatex, bibtex, pdflatex, and pdflatex again to ensure all references are correctly processed.

17. What are the benefits and downsides of using PDFLaTeX, XeLaTeX, and LuaLaTeX?

PDFLaTeX, XeLaTeX, and LuaLaTeX each have unique benefits and downsides. PDFLaTeX is the most compatible with older systems and has a large package repository. However, it lacks Unicode support and struggles with non-Latin scripts.

XeLaTeX offers full Unicode support and handles any font on your system, making it ideal for multilingual documents. It’s slower than PDFLaTeX and some packages are incompatible.

LuaLaTeX combines the strengths of both: compatibility from PDFLaTeX and Unicode/font handling from XeLaTeX. Additionally, it provides scripting capabilities via Lua. Its downside is its speed; it’s the slowest among the three

18. Can you explain the process of managing large LaTeX documents, such as splitting them into separate files?

Managing large LaTeX documents involves splitting them into separate files for better organization and readability. This process is known as modularization.

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To begin, create a main file that will serve as the root document. In this file, use the \documentclass{} command to define the type of document you're creating. Then, include the preamble where you can specify packages needed, custom commands, and formatting rules.

Next, split your content into smaller .tex files based on logical divisions such as chapters or sections. Each of these files should contain only the text of that section, without any preamble or \begin{document} and \end{document} commands.

Then, in your main file, use the \include{} or \input{} commands to incorporate these separate files. The \include{} command starts a new page before the content of the included file, while \input{} simply inserts the content at its current location.

Finally, compile the main file. LaTeX will automatically integrate all the included files into one cohesive document.

19. How do you troubleshoot LaTeX errors, such as undefined references or missing packages?

To troubleshoot LaTeX errors, start by examining the error message. It usually indicates the line number and nature of the problem. For undefined references, ensure that you've correctly labeled your sections, figures, or tables and used the right command to refer them. If it's a missing package error, check if the package is installed in your TeX distribution. If not, install it using the package manager. Sometimes, the issue might be due to outdated packages; updating your TeX distribution can resolve this. In case these steps don't work, online forums like StackExchange provide solutions for specific problems.

20. Can you explain how to use LaTeX to create complex diagrams or graphics?

LaTeX, while primarily used for document preparation, can also be utilized to create complex diagrams or graphics. This is achieved through the use of packages such as TikZ and PGFPlots.

TikZ allows users to generate vector graphics from within LaTeX. It's a powerful tool that enables the creation of line drawings, curves, and plots. To use it, include \usepackage{tikz}

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in your preamble. Then, you can begin creating your diagram within a tikzpicture environment. For example:

```
\begin{tikzpicture}
\draw (0,0) –(1,2);
\end{tikzpicture}
```

This will draw a line from coordinates (0,0) to (1,2).

PGFPlots, on the other hand, is excellent for creating 2D and 3D plots. Include \usepackage{pgfplots} in your preamble to utilize this package. You can then start plotting within an axis environment. For instance:

```
\begin{axis}
\addplot {x^2};
\end{axis}
```

This will plot a simple quadratic function.

Remember, both packages have extensive documentation available online for more detailed instructions and examples.

21. How would you import and format code snippets from other programming languages in a LaTeX document?

To import and format code snippets in LaTeX, use the “listings” package. Include \usepackage{listings} in your preamble to import it. To set default formatting for all listings, use \lstset{} command with desired parameters like language, basicstyle, numbers etc. For example:

```
\lstset{
language=Python,
basicstyle=\small\ttfamily,
numbers=left
}
```


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To include a code snippet from an external file, use \lstinputlisting[language=Python]{file.py}. Replace 'Python' with the programming language of your choice and 'file.py' with the path to your source file.

For inline code, use \lstinline!code!. Replace 'code' with the actual code you want to display.

22. How would you create a table of contents, list of figures, and list of tables in LaTeX?

To create a table of contents in LaTeX, use the command \tableofcontents. This automatically generates a table based on your document's sections and subsections. For a list of figures, use \listoffigures, which creates a list from all labeled figures within your document. Similarly, for a list of tables, use \listoftables to generate a list from all labeled tables. These commands should be placed where you want these lists to appear, typically at the beginning of the document after the abstract or introduction. Remember to compile twice to ensure accurate page numbers.

23. Can you provide your preferred method for making a LaTeX document more readable and easier to maintain?

My preferred method for enhancing readability and maintainability of a LaTeX document involves several strategies. First, I use comments extensively to explain the purpose of complex equations or formatting decisions. This helps future readers understand my thought process. Second, I break up long sections into smaller files using the \input{} command. This reduces clutter in the main file and makes it easier to locate specific sections. Third, I consistently follow naming conventions for labels and references, which simplifies navigation through the document. Fourth, I utilize packages like 'amsmath' for better mathematical notation and 'hyperref' for clickable links within the document. Lastly, I ensure regular indentation and spacing between different elements for visual clarity.

24. How do you handle version control with LaTeX documents?

Version control in LaTeX can be managed using Git, a distributed version control system. The process involves initializing a new repository with 'git init', adding files to the repository with 'git add', and committing changes with 'git commit'. To track changes between versions, use 'git diff'. For collaboration, push your commits to an online repository like

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GitHub with ‘git push’. Colleagues can then clone the repository with ‘git clone’ or pull updates with ‘git pull’. Merge conflicts may occur when two people edit the same part of a document; these are resolved manually. Remember to regularly commit and push changes to avoid large, complex merges.

25. Can you explain how you would use LaTeX for a large project, such as writing a book or a thesis?

LaTeX is ideal for large projects due to its structure and formatting capabilities. For a book or thesis, I would start by creating a main document serving as the project’s backbone. This file will contain the preamble with all necessary packages, custom commands, and document class.

I’d then create separate .tex files for each chapter or section of the work. These files are included in the main document using the \include{} command. This approach keeps the project organized and manageable, allowing focus on one section at a time without scrolling through hundreds of pages.

For bibliography management, BibTeX or Biber with BibLaTeX package can be used. Citations are inserted directly into the text and the reference list is automatically generated and formatted.

To handle complex structures like tables, figures, equations, LaTeX provides environments that ensure correct placement and numbering. Cross-referencing is also simplified with \label{} and \ref{} commands.

Finally, version control systems such as Git can be utilized alongside LaTeX for tracking changes and collaboration purposes.

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